CODEMITVISHAL · FREE COURSE

Python Pro

From Python Foundations to **professional** Python and web development: clean code, OOP, testing, concurrency, FastAPI, Django, security and Azure.

21 modules

60+ runnable examples

0 € always free

COURSE MAP

13 core modules, then an 8-module web track

#	MODULE	YOU'LL BE ABLE TO…
08	**Pythonic code**	write short, idiomatic Python
09	**Data structures deep dive**	pick the right structure (Big-O)
10	**Advanced functions**	write closures and decorators
11	**Iterators & generators**	process data lazily
12	**Advanced OOP**	design clean classes (SOLID)
13	**Errors, context managers, logging**	fail safely, log clearly
14	**Project structure & packaging**	ship an installable CLI

#	MODULE	YOU'LL BE ABLE TO…
15	**Type hints & code quality**	catch bugs before running
16	**Testing with pytest**	prove your code works
17	**Concurrency**	use threads, processes, asyncio
18	**Data, APIs & databases**	build a REST API with a DB
19	**Performance & internals**	profile and speed things up
20	**Capstone project**	build it all, end to end

Every module: slides · runnable code · exercise + solution · quiz

MODULE

08

Pythonic code

Same results, half the code. Write Python the way experienced Python developers do.

IN THIS MODULE

Comprehensions	`01_comprehensions.py`
Unpacking	`02_unpacking.py`
Looping like a pro	`03_looping_like_a_pro.py`
Truthiness & the walrus :=	`04_truthiness_and_walrus.py`
Strings like a pro	`05_strings_like_a_pro.py`
Demo: grade report	`06_demo_grade_report.py`
Exercise: refactor to Pythonic	`ex08_refactor_to_pythonic.py`

LESSON · COMPREHENSIONS

Build a list in one readable line

```
[ p * 1.19 for p in prices if p < 20 ]
```

① what to keep ② the loop ③ optional filter

```
# the old way: 4 lines
with_tax = []
for p in prices:
    with_tax.append(p * 1.19)

# Pythonic: 1 line
with_tax = [p * 1.19 for p in prices]
```

KIND	SYNTAX
list	`[x for x in xs]`
dict	`{k: v for k, v in pairs}`
set	`{x.lower() for x in xs}`
generator	`sum(x for x in xs)`

> **if/else before for** chooses a value. **if after for** filters.

LESSON · UNPACKING

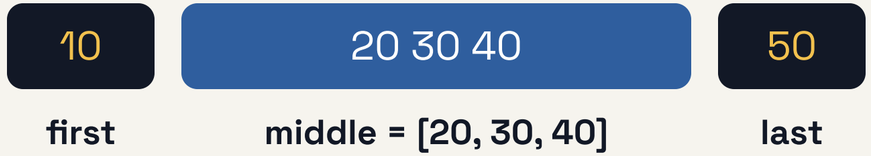
Pull values out and into names, in one step

```
x, y = (3, 7)                  # basic
a, b = b, a                    # swap, no temp
first, *middle, last = nums    # * = "the rest"
name, _, city = record         # _ = ignore

area(*[4, 5])                  # list -> args
area(**{"width": 2, "height": 9})

merged = [*odds, *evens]
settings = defaults | user     # later keys win
```

first, *middle, last = [10, 20, 30, 40, 50]



> The number of names must match the number of values, **unless** one name has a `*`.

> `*` spreads a list into arguments, `**` spreads a dict into keyword arguments.

LESSON · LOOPING LIKE A PRO

Stop writing range(len(...))

INSTEAD OF…	WRITE
`for i in range(len(xs)): xs[i]`	`for i, x in enumerate(xs, start=1)`
Two lists with one index	`for a, b in zip(xs, ys)`
A custom sort function	`sorted(xs, key=len, reverse=True)`
A flag + loop + break	`any(x < 0 for x in xs)`
”Is every item OK?” loop	`all(x >= 50 for x in xs)`
Track the biggest by hand	`max(pairs, key=lambda p: p[1])`
`for k in d: d[k]`	`for k, v in d.items()`

```
for s in scores:
    if s < 50:
        print("fail found")
        break
else:
    print("all passed")
```

> **for … else:** the `else` runs only if the loop finished **without** `break`.

> `zip` stops at the shortest list. Add `strict=True` to catch length bugs.

LESSON · TRUTHINESS & WALRUS

Short, clear conditions

FALSY

`0 0.0 "" [] {} set() None False`

Everything else is truthy, even `"0"` and `[0]`.

INSTEAD OF…	WRITE
`if len(cart) == 0:`	`if not cart:`
`if x == None:`	`if x is None:`
`x >= 50 and x < 80`	`50 <= x < 80`
if/else to pick a default	`name or "Guest"`

```
# walrus := assigns AND returns
while (line := input()) != "quit":
    print(line.upper())

if (n := len(data)) > 3:
    print(f"long list: {n} items")
```

⚠️ `count or 10` turns a real **0** into 10, because 0 is falsy. Use `10 if count is None else count`.

LESSON · STRINGS LIKE A PRO

f-strings format, align and even debug

SPEC	EXAMPLE	RESULT
`.2f`	`f"{1234.5678:.2f}"`	**1234.57**
`,.2f`	`f"{1234.5:,.2f}"`	**1,234.50**
`.1%`	`f"{0.4567:.1%}"`	**45.7%**
`05d`	`f"{42:05d}"`	**00042**
`<10 / >10`	`f"{name:<10}|"`	**left / right align**
`=`	`f"{user=}"`	**user='Vishal'**

```
" ".join(["learn", "python"])
# 'learn python'

name, age, city = line.split(",")
# split + unpack in one go
```

> Strings are **immutable**: every method returns a **new** string.

> Build big text with a list and `"".join()`, not `+=` in a loop.

DEMO · GRADE REPORT

Everything from Module 8 in one program

```
for line in raw.splitlines():
    name, *marks = line.split(",")          # unpacking
    students[name] = [int(m) for m in marks]  # list comp

averages = {n: sum(m) / len(m)                 # dict comp
            for n, m in students.items()}
ranking = sorted(averages.items(),
                 key=lambda p: p[1], reverse=True)

for place, (name, avg) in enumerate(ranking, 1):
    result = "PASS" if avg >= 50 else "FAIL"
    print(f"{place:<3}{name:<10}{avg:>6.1f}  {result}")

print(all(a >= 50 for a in averages.values()))
```

OUTPUT

```
#  Name          Avg  Result
1  Vishal       91.3  PASS
2  Li           85.3  PASS
3  Raj          70.7  PASS
4  Sam          45.0  FAIL

Everyone passed?  False
```

> Run it: `python 08_pythonic_code/06_demo_grade_report.py`

EXERCISE 8 · YOUR TURN

Refactor 9 functions to Pythonic code

The task

> Each function works, but it's written the long way. Rewrite each one with the **Module 8 tool that fits**. Keep the names and results the same.

> `money()` also has a bug: `1234.5` must become **€1,234.50**.

> Run the file: all built-in checks must pass.

comprehension · enumerate · zip · any · max(key=) · or · join

```
# before
result = []
for n in numbers:
    if n % 2 == 0:
        result.append(n * n)

# after
[n * n for n in numbers if n % 2 == 0]
```

Starter:
exercises/ex08_refactor_to_pythonic.py
Solution:
solutions/ex08_refactor_to_pythonic.py

MODULE

09

Data structures deep dive

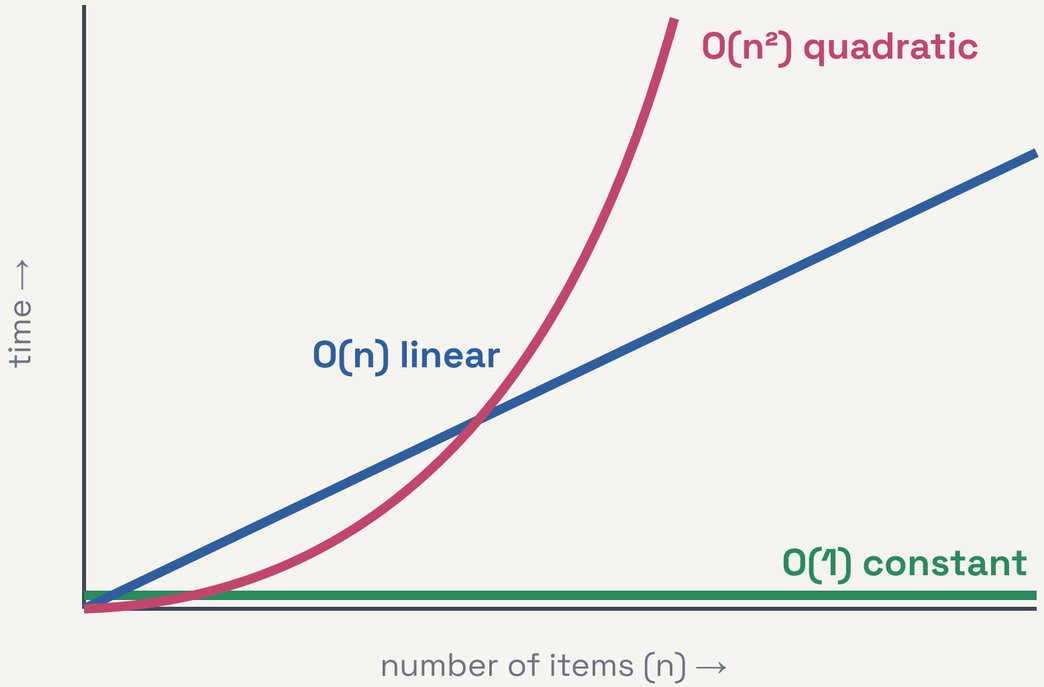
How lists, dicts and sets work inside, and how to pick the right one so your code stays fast.

IN THIS MODULE

Big-O: why it matters	01_big_o.py
Lists & tuples under the hood	02_lists_and_tuples.py
Dicts, sets & hashing	03_dicts_and_sets.py
The collections module	04_collections.py
dataclasses	05_dataclasses.py
Demo: log analyzer	06_demo_log_analyzer.py
Exercise: library tracker	ex09_library_tracker.py

LESSON · BIG-O

How does the work grow when the data grows?



OPERATION	LIST	DICT / SET
`x in …`	O(n)	O(1)
read by index / key	O(1)	O(1)
add at the end	O(1)	O(1)
insert at the front	O(n)	–

MEASURED: "is the last item in here?" × 200

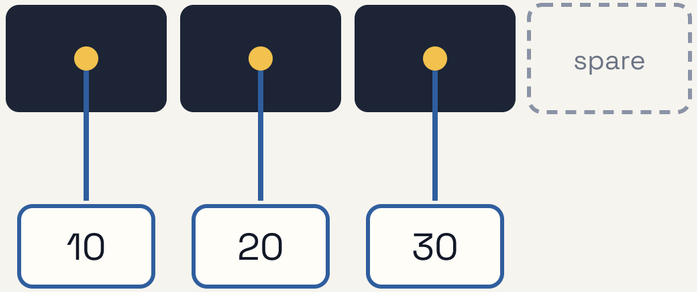
```
    1,000 items | list:    1 ms | set: 0.01 ms
  100,000 items | list:  176 ms | set: 0.02 ms
1,000,000 items | list: 2304 ms | set: 0.01 ms
```

Checking membership a lot? Convert to a **set** once.

LESSON · LISTS & TUPLES UNDER THE HOOD

A list is a row of references, with spare room

a = [10, 20, 30] (capacity 4)



append → spare slot: **fast**

insert(0, x) → shift all: **slow**

each slot = a reference

```
b = a           # same list, 2 names
c = a.copy()    # new outer list (shallow)
d = copy.deepcopy(grid)  # copies inner lists too
```

LIST

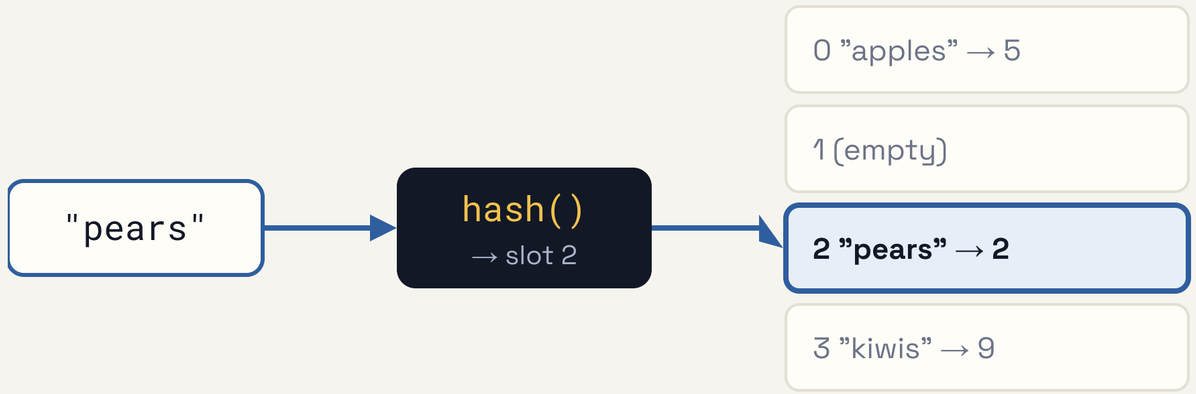
Changeable. Grows and shrinks. Use for **collections of similar items**.

TUPLE

Read-only and lighter (56 vs 72 bytes for 2 items). **Hashable**, so it can be a dict key. Use for **fixed records**: `(x, y)`, RGB, DB rows.

⚠️ A shallow copy shares the inner lists. Nested data? Use **copy.deepcopy**.

Dicts don’t search. They jump straight to the answer.



One calculation, no matter how many keys → **O(1)**

> Keys must be **hashable** = immutable: `str` `int` `tuple` `frozenset`. Not list, dict or set.

SET OP	MEANS
`a | b`	in either (union)
`a & b`	in both (intersection)
`a - b`	only in a (difference)
`a ^ b`	in exactly one

> Dicts keep **insertion order**. Dedupe and keep order: `list(dict.fromkeys(xs))`

LESSON · COLLECTIONS

Four containers that delete boilerplate

COUNTER

Count anything

```
Counter(words).most_common(2)
# [('the', 3), ('and', 2)]
```

DEFAULTDICT

No more "if key not in dict"

```
by_team = defaultdict(list)
by_team[team].append(name)
```

DEQUE

Fast at both ends · keep the last N

```
recent = deque(maxlen=3)
recent.append(page)  # oldest drops off
```

NAMEDTUPLE

A tuple with field names

```
Point = namedtuple("Point", ["x", "y"])
Point(3, 7).x   # 3
```

LESSON · DATACLASSES

Classes for data, without the boilerplate

```
@dataclass
class Product:
    name: str
    price: float
    qty: int = 0                                # default
    tags: list = field(default_factory=list)

    def total(self) -> float:
        return self.price * self.qty

Product("Laptop", 899.0, 2)
# Product(name='Laptop', price=899.0, qty=2, tags=[])
```

YOU GET FOR FREE

__init__ · **__repr__** · **__eq__**, built from the fields.

OPTIONS

`frozen=True` → read-only and hashable
`order=True` → sortable by the fields
`asdict(obj)` → plain dict (great for JSON)

⚠️ Never `tags: list = []`. Every object would share one list. Use **field(default_factory=list)**.

DEMO · LOG ANALYZER

One question, one right data structure

QUESTION	TOOL
One clean record per log line?	**@dataclass Entry**
Most popular pages?	**Counter** → `most_common(3)`
Which pages did each user visit?	**defaultdict(set)**
How many unique visitors?	**set** comprehension
The last 3 server errors?	**deque(maxlen=3)**

OUTPUT

```
Top pages:
  /home  4 hits
  /cart  3 hits
  /shop  2 hits

Unique visitors: 4

Last 3 server errors:
  10:04 /cart (raj)
  10:07 /cart (vishal)
  10:09 /cart (li)
```

EXERCISE 9 · YOUR TURN

Build a library tracker

The task: fill in 6 TODOs

1	Make `Book` a **dataclass** (isbn, title, author)
2	`build_index` : a **dict** isbn → Book (O(1) lookups)
3	`most_borrowed` : top n with **Counter**
4	`titles_by_author` : **defaultdict(list)**
5	`unique_borrowers` : a **set**
6	`ReturnLog` : the last 3 returns with **deque(maxlen=3)**

```
$ python exercises/
    ex09_library_tracker.py
All checks passed!
```

Starter:
exercises/ex09_library_tracker.py
Solution:
solutions/ex09_library_tracker.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

10

Advanced functions

Functions are objects: pass them, return them, make them remember, wrap them.

IN THIS MODULE

Parameters like a pro	01_args_and_kwargs.py
Scope & closures	02_scope_and_closures.py
Functions are objects	03_first_class_functions.py
Decorators	04_decorators.py
functools	05_functools.py
Demo: resilient API client	06_demo_api_client.py
Exercise: write decorators	ex10_decorators.py

LESSON · PARAMETERS

*args, **kwargs, keyword-only and the trap

```
def f(a, b=2, *args, c, d=4, **kwargs):
#     |  |     |      |  |     └ extra keywords -> dict
#     |  |     |      └──┘ keyword-only (after *)
#     |  |     └ extra positionals -> tuple
#     |  └ default
#     └ positional

def connect(host, *, port=443): ...
connect("api.io", port=8080)    # must name it

def add(item, basket=None):     # NOT basket=[]
    if basket is None:
        basket = []
    basket.append(item)
    return basket
```

***args** collects extra positional arguments into a **tuple**, ****kwargs** extra keywords into a **dict**.

Everything after a bare `*` must be passed **by name**: call sites explain themselves.

⚠️ Default values are created **once**. A `[]` default is shared by every call. Use **None**.

A function that remembers

```
def make_counter():
    count = 0
    def next_number():
        nonlocal count      # update the OUTER count
        count += 1
        return count
    return next_number      # the function, not a call

c1, c2 = make_counter(), make_counter()
c1(), c1(), c1(), c2()      # 1 2 3 1
```

> **LEGB**: Python looks for a name in **L**ocal → **E**nclosing → **G**lobal → **B**uilt-in.

> A **closure** keeps the variables of its outer function alive, even after it returned.

> Reading an outer variable just works. **Assigning** needs `nonlocal` (or `global`).

⚠️ `[lambda: i for i in range(3)]` all return 2 (late binding). Fix: **lambda i=i: i**

Wrap a function to add behaviour

```python
def timer(func):
    @functools.wraps(func)          # keep name + docstring
    def wrapper(*args, **kwargs):   # accept ANY arguments
        start = time.perf_counter()
        result = func(*args, **kwargs)
        print(f"{func.__name__}: {time.perf_counter() - start:.3f}s")
        return result               # don't forget!
    return wrapper

@timer                              # same as: slow = timer(slow)
def slow(): ...
```

> **@decorator** above `def f` means `f = decorator(f)`.

> Decorator **with arguments** = one more layer: `retry(times)` → `decorator(func)` → `wrapper(*args)`.

> Stacked decorators: the one **closest to def** wraps first.

LESSON · FUNCTOOLS

Ready-made tools for functions

TOOL	WHAT IT DOES	EXAMPLE
lru_cache	remembers results (memoization)	`fib(200)` : instant
partial	pre-fills arguments	`square = partial(power, exp=2)`
wraps	keeps names in decorators	`@functools.wraps(func)`
reduce	folds a list into one value	`reduce(mul, [1,2,3,4])` → 24
singledispatch	one function, code per type	`describe(21)` vs `describe([1])`

```
@lru_cache(maxsize=None)
def fib(n):
    return n if n < 2 else fib(n-1) + fib(n-2)

fib(28)   # 0.064 s -> 0.00003 s
```

Only cache **pure** functions: same input → same output, no side effects.

EXERCISE 10 · YOUR TURN

Write three decorators

The task

1	`@count_calls` : store the number of calls in `func.calls`
2	`@require_positive` : **ValueError** for negative numbers
3	`@memoize` : your own cache with a dict (no lru_cache!)
4	Work with any arguments and keep the name: **functools.wraps**

```
@count_calls
def hello(): return "hi"

hello(); hello(); hello()
hello.calls        # 3
```

Starter: exercises/ex10_decorators.py
Solution: solutions/ex10_decorators.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

11

Iterators & generators

Process data one item at a time: a 10 GB file with the memory of 10 lines.

IN THIS MODULE

How for really works	`01_iteration_protocol.py`
Generators	`02_generators.py`
Lazy pipelines	`03_pipelines.py`
itertools	`04_itertools.py`
Demo: a 200,000-row CSV	`05_demo_big_csv.py`
Exercise: lazy generators	`ex11_generators.py`

LESSON · THE ITERATION PROTOCOL

What a for loop really does

```
for x in things:           # is really:
    print(x)

it = iter(things)          # iterable -> iterator
while True:
    try:
        x = next(it)       # next item...
    except StopIteration: # ...until there's none
        break
    print(x)
```

Iterable: something you can loop over. It hands out an iterator (`__iter__`).

Iterator: remembers **where you are** and gives the next item (`__next__`).

An iterator is **used up** after one pass: `list(it)` twice gives `[]` the 2nd time.

LESSON · GENERATORS

yield: hand out a value, pause, resume

```
def countdown(start):
    while start > 0:
        yield start        # hand out + PAUSE here
        start -= 1         # resume here on next()

list(countdown(3))         # [3, 2, 1]

squares = (n*n for n in range(10**6))  # generator expression
sum(squares)               # one value at a time
```

> Calling a generator function runs **nothing**; it returns a generator object.

> Each `next()` runs to the next `yield` and keeps all local variables.

> `yield from xs` hands over to another iterable.

Memory: a list of 1M squares = **8.4 MB**, the generator = **200 bytes**.

LESSON · LAZY PIPELINES

Chain generators like an assembly line

```
lines   = read_lines(path)          # yields one line at a time
records = parse(lines)              # yields (date, level, msg)
errors  = only("ERROR", records)    # filters lazily
msgs    = messages(errors)          # formats lazily

for m in islice(msgs, 2):           # take 2, then stop
    print(m)
# only 4 lines were ever read!
```

> Nothing runs until something **asks** for values.

> Each stage does **one job**, so it's easy to test and reuse.

> Demo with a 200,000-row CSV: **56 MB** (list) vs **0.05 MB** (generators).

LESSON · ITERTOOLS

A toolbox of fast, lazy iterators

TOOL	EXAMPLE	RESULT
islice	`islice(count(10, 5), 4)`	10 15 20 25
chain	`chain([1, 2], "ab")`	1 2 'a' 'b'
accumulate	`accumulate([5, 10, 20])`	5 15 35
pairwise	`pairwise([12, 15, 11])`	(12,15) (15,11)
groupby	sort first, then group neighbours	DE 150 · IN 100
product	`product("SM", ["red", "blue"])`	every combination
combinations	`combinations(names, 2)`	every pair

⚠️ **groupby** only groups **consecutive** items. Sort by the same key first.

EXERCISE 11 · YOUR TURN

Write lazy generators

The task

1	`chunked(items, size)` : lists of `size` items
2	`running_average(numbers)` : the average so far
3	`valid_numbers(lines)` : ints only, skip the rest
4	Use **yield**; must work on **endless** input

```
list(chunked("abcde", 2))
# [['a','b'], ['c','d'], ['e']]

list(running_average([2, 4, 6]))
# [2.0, 3.0, 4.0]
```

Starter: exercises/ex11_generators.py
Solution: solutions/ex11_generators.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

12

Advanced OOP

Classes that feel built-in, and designs that stay easy to change.

IN THIS MODULE

Dunder methods	`01_dunder_methods.py`
Properties	`02_properties.py`
Class & static methods	`03_class_and_static_methods.py`
Inheritance & the MRO	`04_inheritance_and_mro.py`
ABCs & Protocols	`05_abc_and_protocols.py`
Composition & SOLID	`06_composition_and_solid.py`
Exercise: a Vector class	`ex12_vector.py`

Python calls them for you

YOU WRITE	PYTHON CALLS
`print(x)` / `repr(x)`	`__str__` / `__repr__`
`a == b`, `a < b`	`__eq__`, `__lt__` (+ @total_ordering)
`a + b`, `2 * v`	`__add__`, `__rmul__`
`len(x)`, `x[i]`, `for i in x`	`__len__`, `__getitem__`
`item in x`	`__contains__`
`{x}`, dict key	`__hash__`
`if x:`	`__bool__`

```
Money(10.5) + Money(4.25)
# 14.75 EUR
sorted([Money(9), Money(1)])
```

> Can't compare with that type? Return **NotImplemented**, not False.

Defined **__eq__**? Define **__hash__** too, or the objects can't go in sets.

Attributes with rules, alternative constructors

```python
class Temperature:
    @property
    def celsius(self):
        return self._celsius
    @celsius.setter
    def celsius(self, value):
        if value < -273.15:
            raise ValueError("below absolute zero!")
        self._celsius = value

class Employee:
    @classmethod
    def from_csv(cls, line):       # cls = the class it's called on
        name, salary = line.split(",")
        return cls(name, int(salary))
```

@property: looks like an attribute, runs a method: validation, computed or read-only values.

@classmethod: alternative constructors. `Manager.from_csv()` returns a **Manager**.

@staticmethod: a plain helper that lives in the class.

One clear order, explicit contracts

```python
class Child(Left, Right): ...
Child.__mro__  # Child, Left, Right, Base, object

class Shape(ABC):                  # nominal: must inherit
    @abstractmethod
    def area(self) -> float: ...

class Notifier(Protocol):          # structural: any class
    def send(self, msg: str) -> None: ...   # with .send() fits
```

super() calls the **next class in the MRO**, not simply "my parent". That's what makes mixins work.

ABC: subclasses that forget an abstract method can't even be created.

Protocol: duck typing with type checking. Great for classes you don't own.

LESSON · COMPOSITION & SOLID

Build from parts, inject the details

```
class Checkout:
    def __init__(self, discount: Discount, notifier: Notifier):
        self.discount = discount        # HAS-A, injected
        self.notifier = notifier

    def pay(self, items):
        total = self.discount.apply(sum(p for _, p in items))
        self.notifier.send(f"charged {total:.2f}")

Checkout(PercentOff(20), ConsoleNotifier()).pay(cart)
Checkout(NoDiscount(), FakeNotifier()).pay(cart)  # in tests
```

Single responsibility · **O**pen for extension · **L**iskov · **I**nterface segregation · **D**ependency inversion

IS-A → inheritance. **HAS-A → composition.**

New discount rule? A **new class**. Checkout never changes.

EXERCISE 12 · YOUR TURN

A Vector class that feels built-in

The task

1	`repr`, `+`, `v * 2` and `2 * v`
2	`abs(v)`, a read-only **length** property
3	`==` and **hashable** (works in sets)
4	`x, y = v`, `bool()`, `Vector.from_tuple()`

```
v = Vector(3, 4)
v + Vector(1, 1)   # Vector(4, 5)
2 * v              # Vector(6, 8)
abs(v)             # 5.0
x, y = v
```

Starter: exercises/ex12_vector.py
Solution: solutions/ex12_vector.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

13

Errors, context managers & logging

Fail safely, clean up always, leave a clear trail.

IN THIS MODULE

Exceptions like a pro	`01_exception_hierarchy.py`
Context managers	`02_context_managers.py`
logging	`03_logging.py`
Demo: robust order importer	`04_demo_order_importer.py`
Exercise: safe bank transfers	`ex13_bank_transfer.py`

Your own exception family

```python
class ShopError(Exception): ...          # the family
class OutOfStock(ShopError):
    def __init__(self, item, left):
        super().__init__(f"only {left} x {item} left")
        self.item, self.left = item, left   # extra data

try:
    buy("laptop", 5)
except OutOfStock as e:      # most specific FIRST
    print(e.left)
except ShopError as e:       # then the family
    ...

raise ShopError("config broken") from err   # keep the cause
```

> Catch as **specific** as possible; a parent catches all its children.

> `raise ... from e` keeps the original error in the traceback.

> **EAFP**: try it, handle the exception. The Pythonic default.

⚠️ Never `except: pass`. It also catches Ctrl+C and hides bugs forever.

Setup and clean-up, guaranteed

```
@contextmanager
def transaction(db):
    snapshot = dict(db)
    try:
        yield                   # the with-block runs here
    except Exception:
        db.clear(); db.update(snapshot)   # roll back
        raise                   # never swallow the error

with transaction(accounts):
    accounts["raj"] += 30
    withdraw(accounts, "vishal", 30)
```

`with x:` → `__enter__`, the body, then `__exit__`, which **always** runs.

Code before `yield` = enter, after = exit.

Files, locks, DB transactions, timers, temp folders, `suppress(FileNotFoundError)`

LESSON · LOGGING

print() for grown-ups

```
logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s %(levelname)-8s %(name)s: %(message)s")
log = logging.getLogger(__name__)    # one per module

log.debug("hidden below INFO")
log.info("order %s created", 1042)   # %s, not f-strings
log.warning("stock low: %d left", 2)
try:
    1 / 0
except ZeroDivisionError:
    log.exception("discount failed")  # + traceback
```

> Levels: **DEBUG < INFO < WARNING < ERROR < CRITICAL**.

> Configure **once** at start-up; add handlers for files or the cloud.

> `%s` arguments: the text is only built if the message is shown.

EXERCISE 13 · YOUR TURN

Safe bank transfers

The task

1	`InsufficientFunds(BankError)` with `.account .balance .amount`
2	`withdraw()` raises it when the balance is too low
3	`atomic(accounts)` : snapshot, roll back, **re-raise**
4	`transfer()` : no money may appear or disappear

```
transfer(accounts, "raj", "vishal", 500)
# InsufficientFunds: raj has 80
accounts   # unchanged: rolled back
```

Starter: exercises/ex13_bank_transfer.py
Solution: solutions/ex13_bank_transfer.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

14

Project structure & packaging

From scripts to an installable project with its own command.

IN THIS MODULE

Modules & imports	`01_modules_and_imports.py`
CLIs with argparse	`02_cli_with_argparse.py`
A real package: tasky	`tasky_project/`
Exercise: wordtool CLI	`ex14_wordtool_cli.py`

What import really does

```
import json
# 1. already in sys.modules? reuse it (runs only ONCE)
# 2. otherwise search the folders in sys.path
# 3. run the file, store it in sys.modules

if __name__ == "__main__":   # only when run directly,
    main()                   # not when imported (tests!)

from tasky.store import TaskStore   # absolute: preferred
from .store import TaskStore        # relative: inside a package
```

module = one .py file · **package** = a folder of modules with `__init__.py`

`__name__` is `"__main__"` only for the file you ran.

Avoid `from x import *`: you can't tell where names come from.

LESSON · A REAL PACKAGE

src layout + pyproject.toml

```
tasky_project/
├── pyproject.toml      # name, version, deps, commands
├── src/tasky/
│   ├── __init__.py     # public API, __version__
│   ├── __main__.py     # python -m tasky
│   ├── cli.py          # argparse -> calls store
│   └── store.py        # the logic (no printing)
└── tests/test_store.py

[project.scripts]
tasky = "tasky.cli:main"     # creates the `tasky` command
```

`pip install -e .`: an **editable** install; code changes apply instantly.

Keep **logic** (store.py) apart from the **interface** (cli.py): reusable and testable.

Return a **non-zero exit code** on failure so scripts and CI can tell.

$ tasky add "Record Module 14" · $ tasky done 1 · $ python -m tasky list

EXERCISE 14 · YOUR TURN

Build a CLI with argparse

The task

1	Subcommand `stats TEXT` : words, unique, chars
2	Subcommand `top TEXT --n 3` : the most common words
3	`main(argv=None)` : testable without a terminal
4	Case-insensitive counting; return exit code 0

```
$ wordtool stats "the cat and the hat"
words: 5, unique: 4, chars: 19

$ wordtool top "the cat and the hat" --n 1
the: 2
```

Starter: exercises/ex14_wordtool_cli.py
Solution: solutions/ex14_wordtool_cli.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

15

Type hints & code quality

Catch bugs before you run anything, and keep code easy to read.

IN THIS MODULE

Type hints	01_type_hints_basics.py
Advanced typing	02_advanced_typing.py
mypy catches bugs	03_mypy_catches_bugs.py
ruff & style	04_ruff_and_style.py
Exercise: type it, fix it	ex15_add_type_hints.py

LESSON · TYPE HINTS

Say what your code expects

TYPE	MEANS
`list[int]`, `dict[str, float]`	what's **inside** a collection
`float | None`	a float, **or None** (handle it!)
`Literal["info", "error"]`	only these exact values
`TypedDict`	a dict with known keys (JSON)
`Callable[[float], float]`	a function float → float
`TypeVar T`: `list[T] -> T`	generic: str in, str out
`Final`	a constant

```
def find_price(item: str) -> float | None:
    return prices.get(item)
```

> Python does **not** check hints at runtime.
> Your editor and **mypy** do.

LESSON · MYPY

3 hidden crashes, found in one second

```python
def plan_of(user_id: int) -> str:
    user = find_user(user_id)         # dict | None
    return user.get("plan", "free")  # crashes for unknown ids

def label(count: int) -> str:
    if count > 100:
        return "Orders: " + count    # str + int

def bill(total: float) -> None:
    send_invoice(total)              # expects an email str
```

$ mypy 03_mypy_catches_bugs.py

> `Item "None" ... has no attribute "get"`

> `Unsupported operand types for + ("str" and "int")`

> `Argument 1 ... incompatible type "float"; expected "str"`

The code runs fine with today's test data. mypy finds the crashes **before** production does.

LESSON · RUFF

Lint for bugs, format for consistency

```
import os, sys                          # E401 + F401 unused
def messy_total( items ,discount = None ):
    for i in range(len(items)):
        if items[i]["price"] == None:   # E711
            continue
    try: ...
    except:                             # E722 bare except
        return 0

$ ruff check file.py      # list problems
$ ruff check --fix        # fix the safe ones
$ ruff format             # rewrite the layout
```

> **Linter**: likely bugs and bad patterns. **Formatter**: layout only.

> Configure once in `pyproject.toml` or `ruff.toml`.

> Tools can't name things for you: `unpaid_invoices`, not `lst2`.

EXERCISE 15 · YOUR TURN

Type it, let mypy find the bug

The task

1. Add hints to every function, with a **TypedDict** for orders
2. Run `mypy --strict`
3. Fix the real bug mypy reports
4. Checks AND mypy --strict must pass

```
$ mypy --strict ex15_add_type_hints.py
error: Value of type "Order | None"
       is not indexable
# report([]) would crash!
```

Starter:
exercises/ex15_add_type_hints.py
Solution:
solutions/ex15_add_type_hints.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

16

Testing with pytest

Prove your code works, and keep it working when it changes.

IN THIS MODULE

pytest basics	`test_01_basics.py`
Fixtures & parametrize	`test_02_fixtures_and_parametrize.py`
Fakes, mocks & monkeypatch	`test_03_mocking.py`
TDD	`test_04_tdd.py`
Exercise: tests that catch bugs	`ex16_test_password.py`

Plain assert, nothing else to learn

```python
def test_discount_takes_percentage_off():
    price = 100.0                          # arrange
    result = apply_discount(price, 20)     # act
    assert result == 80.0                  # assert

def test_over_100_is_rejected():
    with pytest.raises(ValueError, match="between 0 and 100"):
        apply_discount(10, 150)

def test_floats():
    assert 0.1 + 0.2 == pytest.approx(0.3)
```

> pytest runs files `test_*.py` and functions `test_*`.

> A failed assert shows **both values**: `assert 6.0 == 7`.

> Test the **edges**: 0, empty, the limit, invalid input.

$ pytest 16_testing -v → 31 passed

Reusable setup, many cases

```python
@pytest.fixture                          # in conftest.py: shared
def sample_cart():
    return {"book": (12.5, 2), "pen": (1.2, 3)}

def test_total(sample_cart):             # asked for by NAME
    assert cart_total(sample_cart) == pytest.approx(28.6)

@pytest.mark.parametrize(("price", "pct", "expected"), [
    (100, 10, 90.0),
    (19.99, 50, 9.99),                  # floats!
])
def test_cases(price, pct, expected):
    assert apply_discount(price, pct) == expected
```

A fixture is built **fresh for every test**, so tests can't affect each other.

`tmp_path` : a built-in fixture, a temp folder per test.

Each parameter set is reported as its **own** test.

LESSON · MOCKS & TDD

No network in unit tests · red → green → refactor

```python
def describe_weather(city, fetch=fetch_json): ...

def test_hot_city():
    fake = Mock(return_value={"temp_c": 31.4})
    assert describe_weather("Mumbai", fetch=fake) == "Mumbai: 31°C, hot"
    fake.assert_called_once()

def test_api_down():
    fake = Mock(side_effect=TimeoutError("down"))
    with pytest.raises(TimeoutError):
        describe_weather("Oslo", fetch=fake)
```

> Real APIs are slow, need a network, and change: tests become **flaky**.

> **Inject** the dependency, or `monkeypatch` it for one test.

> **TDD**: write a failing test → make it pass → refactor. `slugify()` was built that way.

EXERCISE 16 · YOUR TURN

Write tests that catch bugs

The task

1	`is_strong_password()` is correct: test every rule
2	8+ chars · upper · lower · digit · no spaces · TypeError
3	Each test breaks **only one** rule
4	Your tests must catch **6/6** planted bugs

```
$ python ex16_test_password.py
Your tests caught 6/6 bugs.
All checks passed!
```

Starter:
exercises/ex16_test_password.py
Solution:
solutions/ex16_test_password.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

17

Concurrency

Threads for waiting, processes for calculating, asyncio for thousands of tasks.

IN THIS MODULE

Threads	`01_threads.py`
Processes	`02_processes.py`
asyncio	`03_asyncio_basics.py`
Demo: price checker	`04_demo_price_checker.py`
Exercise: parallel downloads	`ex17_parallel_downloads.py`

LESSON · CHOOSING

Waiting or calculating?

WORK	EXAMPLE	USE	MEASURED
I/O-bound (waiting)	downloads, DB, files	**threads** or **asyncio**	4.0 s → **0.5 s**
CPU-bound (calculating)	maths, images, parsing	**processes**	5.1 s → **2.1 s**
CPU-bound with threads		❌ the GIL	5.1 s → **6.1 s**

```
with ThreadPoolExecutor(8) as pool:
    results = list(pool.map(download, files))

if __name__ == "__main__":     # required!
    with ProcessPoolExecutor() as pool:
        totals = list(pool.map(count_primes, jobs))
```

> The **GIL** lets one thread run Python at a time per process. Separate processes each have their own.

Shared data needs a lock

```
counter = 0
lock = threading.Lock()

def safe_add(n):
    global counter
    for _ in range(n):
        with lock:              # one thread at a time
            counter += 1

# 4 threads x 1000:
#   without the lock: 1204   <- updates lost!
#   with the lock:    4000
```

A **race condition**: two threads read, both add 1, both write. One update is lost.

`with lock:` makes the read-modify-write happen as one step.

Simplest of all: **don't share**. Return results from the pool instead.

LESSON · ASYNCIO

Thousands of waiting tasks on one thread

```python
async def fetch(name, s):
    await asyncio.sleep(s)        # pause, let others run
    return name

async def main():
    results = await asyncio.gather(fetch("x", 1), fetch("y", 1))   # ~1 s
    await asyncio.wait_for(fetch("slow", 5), timeout=0.5)         # TimeoutError
    limit = asyncio.Semaphore(10)                                 # max 10 at once

asyncio.run(main())
```

> Switches happen **only at await**, so race conditions are rarer.

> `gather` = run concurrently · `wait_for` = timeout · `Semaphore` = limit.

⚠️ One blocking call (`time.sleep`, `requests.get`) freezes the whole loop.

EXERCISE 17 · YOUR TURN

Parallel downloads

The task

1	10 downloads × 0.3 s in **< 1 s** with a **ThreadPoolExecutor**
2	The same with **asyncio.gather**
3	Count words with a **ProcessPoolExecutor**
4	Keep the results in input order

```
$ python ex17_parallel_downloads.py
threads: 0.31 s, asyncio: 0.31 s
All checks passed!
```

Starter:
exercises/ex17_parallel_downloads.py
Solution:
solutions/ex17_parallel_downloads.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

18

Data, APIs & databases

Files, web APIs, SQL, and your own REST API with FastAPI.

IN THIS MODULE

Files, CSV & JSON	`01_files_csv_json.py`
Calling web APIs	`02_http_requests.py`
SQLite	`03_sqlite.py`
Your own REST API	`04_fastapi_app.py`
Exercise: a notes database	`ex18_notes_repository.py`

LESSON · WEB APIS

Call an API safely

```
try:
    resp = requests.get(
        "https://api.github.com/search/repositories",
        params={"q": "language:python", "per_page": 3},
        timeout=10)                    # ALWAYS
    resp.raise_for_status()            # 4xx/5xx -> error
    data = resp.json()
except requests.Timeout:
    ...
except requests.RequestException:   # offline, DNS, HTTP
    ...
```

> `params=` encodes query values correctly; never glue strings.

> Tokens live in **environment variables**, never in the code.

> `pathlib` + `csv.DictReader` + `json` for files: lesson 01.

LESSON · SQLITE

SQL with ? parameters and transactions

```
conn.execute("SELECT * FROM products WHERE name = ?", (name,))

user_input = "x' OR '1'='1"
f"... WHERE name = '{user_input}'"   # matches 3 rows: INJECTION
"... WHERE name = ?", (user_input,)  # matches 0 rows: safe

with conn:                                # all or nothing
    conn.execute("UPDATE products SET stock = stock - 1 ...")
    conn.execute("INSERT ... duplicate name")   # IntegrityError
# -> the UPDATE was rolled back too
```

> SQLite: a full SQL database in **one file**, built into Python.

> `conn.row_factory = sqlite3.Row` → `row["name"]`.

> The same ideas work for PostgreSQL and MySQL.

LESSON · FASTAPI

Type hints become a validated web API

```
class TodoIn(BaseModel):
    title: str = Field(min_length=1, max_length=100)

@app.post("/todos", status_code=201)
def create_todo(todo: TodoIn) -> Todo: ...

@app.get("/todos/{todo_id}")
def read_todo(todo_id: int) -> Todo:   # 404 if missing
    ...

# POST {"title": ""}  -> 422 (validated for you)
# $ uvicorn 04_fastapi_app:app --reload  -> /docs
```

pydantic models validate the JSON body automatically.

Status codes: **201** created · **204** no content · **404** not found · **422** invalid.

`TestClient` calls the API in tests without a server.

EXERCISE 18 · YOUR TURN

A notes database

The task

1	`NotesRepository(path)` creates the table
2	`add / get / search / delete / count`
3	Tags stored as **JSON** text
4	Only **?** parameters: the injection check must pass

```
repo = NotesRepository(":memory:")
nid = repo.add("Shopping", "milk", tags=["home"])
repo.search("MILK")    # case-insensitive
repo.search("x' OR '1'='1")   # []
```

Starter:
exercises/ex18_notes_repository.py
Solution:
solutions/ex18_notes_repository.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

19

Performance & internals

Understand what Python does under the hood, then measure before you optimise.

IN THIS MODULE

The memory model	`01_memory_model.py`
Measure first	`02_measure_first.py`
Speed tricks that matter	`03_speed_tricks.py`
Memory tricks	`04_memory_tricks.py`
Exercise: make it fast	`ex19_make_it_fast.py`

LESSON · THE MEMORY MODEL

Names point to objects

```
a = [1, 2, 3]
b = a               # 2 names, 1 object
b.append(4)         # a sees it too

x = 10; y = x
y += 1              # ints are immutable: y -> NEW object

def add(lst): lst.append(1)   # the object is passed,
add(a)                         # so the caller's list changes

int("1000") is int("1000")     # False: use == for values
```

> Every object has an **identity** (id), a **type** and a **value**.

> **Reference counting**: freed the moment the count hits 0; `gc` cleans up cycles.

> Use `is` only for `None`, `True`, `False`.

LESSON · SPEED

Measure first, then fix the biggest thing

FIX	MEASURED SPEED-UP
Better data structure: list `in` → set/dict	**309×**
Cache a slow pure function: `@lru_cache`	**91×**
`"".join()` instead of `+=` in a loop	**22×**
Built-ins (`sum`, `sorted`, `max`) instead of loops	**4×**
Compute once, outside the loop	**3×**
`__slots__` on millions of objects	24 → 17.6 MB

```
$ python -m cProfile -s cumtime app.py

ncalls  cumtime  function
     1    0.222  report
     1    0.172  slow_unique_customers  <- here!
```

> **timeit** compares snippets · **cProfile** finds slow functions · **tracemalloc** finds memory-hungry lines.

EXERCISE 19 · YOUR TURN

Make it fast

The task

1	`find_duplicates` : O(n²) → O(n) with **Counter**
2	`common_customers` : use **set** intersection
3	`fib(300)` : stop recomputing with **lru_cache**
4	Each must fit a **0.1 s** budget

```
# before
find_duplicates(big)   # 1.28 s
# after
find_duplicates(big)   # 0.004 s
```

Starter: exercises/ex19_make_it_fast.py
Solution: solutions/ex19_make_it_fast.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

20

Capstone project

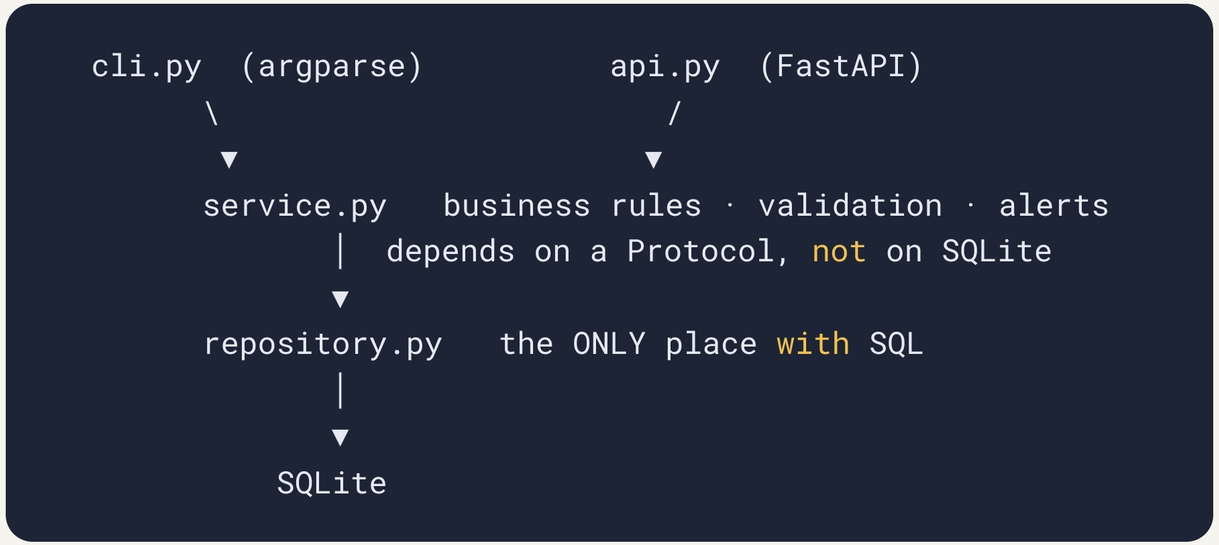
budget: a tested, typed, packaged expense tracker with a CLI and a REST API.

IN THIS MODULE

The project	`budget_project/`
Architecture	`budget_project/README.md`
Tests	`budget_project/tests/`
Challenges	`ex20_capstone_challenges.md`
Final exam	`20_capstone/QUIZ.md`

CAPSTONE · ARCHITECTURE

Clean layers: each talks only to the one below



> The **service** doesn't know whether it's called from a terminal or over HTTP.

> Tests inject an **in-memory** repository: fast and isolated.

> `pytest` · `mypy --strict` · `ruff`: all green.

CAPSTONE · WHERE EVERY MODULE SHOWS UP

Everything you learned, in one project

MODULE	IN BUDGET
08 Pythonic · 09 Data structures	comprehensions, `sorted(key=)`, frozen dataclass, defaultdict
10 Functions · 12 OOP	FastAPI dependencies, Protocol + dependency injection
13 Errors & logging	`BudgetError`, `raise ... from`, `getLogger(__name__)`
14 Packaging	src layout, `pyproject.toml`, the `budget` command
15 Typing · 16 Testing	mypy --strict, ruff, 9 tests with fixtures + TestClient
18 Data & APIs	SQLite repository, FastAPI with pydantic

```
$ budget add 320 food "groceries"
$ budget summary --month 2026-11
2026-11: total 1315.50
  rent    900.00
  food    415.50  <- over budget!
```

EXERCISE 20 · YOUR TURN

Capstone challenges

The task

1	⭐	`budget list` and a transport limit
2	⭐⭐	CSV export · filtered `GET /expenses` · limits from JSON
3	⭐⭐⭐	recurring expenses (generators) · currency + lru_cache
4	⭐⭐⭐	a second repository behind the same Protocol

```
# for EVERY challenge:
# 1. write the test first
# 2. make it pass
# 3. keep pytest, mypy, ruff green
```

Starter:
exercises/ex20_capstone_challenges.md

> Done? Open a **Pull Request** and I'll review your code.

match / case: match the SHAPE of data

```
match event:
    case {"type": "order", "items": [first, *rest]}:
        return f"order: {first} (+{len(rest)} more)"
    case {"type": "refund", "amount": float(a)} if a > 0:
        return f"refund {a:.2f}"
    case Point(x=0, y=0):
        return "the origin"
    case _:
        return "something else"
```

> Match **literals**, **sequences**, **dicts** (extra keys ignored) and **classes**.

> `if ...` after a pattern is a **guard**.

> A bare name **captures**; compare constants with `Color.RED`.

Also new in Module 8: **08_regex.py**, named groups, `re.sub`, validation

The right tool for constants, priorities and sorted data

TOOL	USE IT FOR	EXAMPLE
StrEnum	fixed string options (JSON/DB friendly)	`OrderStatus.PAID == "paid"`
IntEnum / Flag	sortable levels / combinable permissions	`READ | WRITE`
heapq	priority queue, top-k	`heappop(jobs)` → most urgent
heapq.nlargest	top 3 of 1M without a full sort	O(n log k)
bisect	search/insert in sorted lists	`insort(timeline, 30)`

Typos in plain strings fail silently. **Enum typos fail loudly.**

BONUS · MODULES 11-13

Deeper language features

LESSON	KEY IDEA
gen.send(value)	two-way generators: `value = yield result`
async generators	`async for item in all_items()` : stream API pages
descriptors	`__get__/__set__/__set_name__` : how `@property` works
__new__ / __init_subclass__	control creation; auto-register plugins
metaclasses	the class of a class: framework territory
ExitStack	a variable number of context managers, all cleaned up
ExceptionGroup + except*	report and handle several errors at once

BONUS · MODULES 14-19

Pro tooling and performance

LESSON	KEY IDEA
PyPI, uv, pre-commit	`python -m build` · `uv add` · checks on every commit
Generic[T], ParamSpec, @overload	typed containers and decorators that keep signatures
hypothesis + coverage	properties tested on hundreds of generated inputs; `--cov`
TaskGroup + Queue	structured concurrency, backpressure, `httpx.AsyncClient`
dates & zoneinfo	store UTC, show local; DST makes a day 23 hours
pickle vs JSON	loading an untrusted pickle can run code
dis, weakref, NumPy	bytecode, weak caches, 28× faster vectorised maths

WEB DEVELOPMENT TRACK

From HTTP basics to a deployed, secure API

#	MODULE	YOU'LL BE ABLE TO…
21	**HTTP, Flask & Jinja2**	understand HTTP; build server-rendered pages
22	**FastAPI in depth**	design production APIs with validation and DI
23	**SQLAlchemy 2.0 & Alembic**	use an ORM, avoid N+1, migrate schemas
24	**Authentication & security**	hash passwords, issue JWTs, stop OWASP bugs
25	**Django + DRF**	build Django apps with an admin and REST API
26	**WebSockets, tasks & caching**	push live updates, run jobs, cache with Redis
27	**Docker, CI/CD & Azure**	containerise, test in CI, deploy to Azure
28	**Web capstone: taskboard**	ship a secure, tested, real-time API

MODULE

21

HTTP, Flask & Jinja2

How the web really works, then server-rendered pages with Flask.

IN THIS MODULE

HTTP from scratch	`01_http_from_scratch.py`
Flask basics	`02_flask_basics.py`
Templates, forms & sessions	`03_templates_forms_sessions.py`
Exercise: guestbook	`ex21_flask_guestbook.py`

LESSON · HTTP

HTTP is just text

```
GET /hello?name=Vishal HTTP/1.1      <- request line
Host: localhost:8765                 <- headers
Accept: application/json
                                     <- blank line ends the headers
HTTP/1.0 200 OK                      <- status line
Content-Type: application/json
Set-Cookie: visits=1

{"message": "Hello, Vishal!"}        <- body
```

2xx success · **3xx** redirect · **4xx** your fault · **5xx** server fault

201 created · **204** no content · **401** who are you? · **403** not allowed · **422** invalid

Cookies: the server asks the browser to remember a value and send it back.

URLs → functions

```
@app.get("/books/<int:book_id>")
def get_book(book_id):
    book = BOOKS.get(book_id)
    if book is None:
        abort(404, description="not found")
    return book                          # dict -> JSON

@app.post("/books")
def create_book():
    data = request.get_json(silent=True) or {}
    if not data.get("title"):
        return {"error": "title is required"}, 400
    return new_book, 201
```

> `request.args` query · `request.form` forms · `request.get_json()` JSON

> `app.test_client()` tests without a server.

Server-rendered pages, safely

```
{% extends "base.html" %}
{% block content %}
  {% for book in books %}
    <li>{{ book.title|title }}: {{ "%.2f"|format(book.price) }}</li>
  {% endfor %}
  <p>{{ review }}</p>     {# <script> becomes &lt;script&gt; #}
{% endblock %}

# POST -> redirect -> GET
flash("Added!"); return redirect(url_for("shop"))
```

Auto-escaping blocks XSS. Never `|safe` user input.

Redirect after POST: a refresh doesn't resubmit.

session = a signed cookie; `secret_key` from an env var.

EXERCISE 21 · YOUR TURN

A Flask guestbook

The task

1	GET / renders entries, newest first
2	POST /sign validates + flashes + redirects
3	Escape user messages (no |safe)
4	GET /api/entries?name= returns JSON

```
POST /sign  -> 302 /
"Thanks for signing!"
GET /api/entries?name=RAJ
[{"name": "Raj", ...}]
```

Starter:
exercises/ex21_flask_guestbook.py
Solution:
solutions/ex21_flask_guestbook.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

22

FastAPI in depth

Production API design: validation, dependencies, errors, middleware, settings.

IN THIS MODULE

pydantic models	`01_pydantic_models.py`
Routers & dependencies	`02_routers_and_dependencies.py`
Errors, middleware & CORS	`03_errors_middleware_cors.py`
Settings, lifespan & async	`04_settings_lifespan_async.py`
Exercise: library API	`ex22_fastapi_library_api.py`

Validate at the edges

```python
class UserIn(BaseModel):
    username: str = Field(min_length=3, pattern=r"^[a-z0-9_]+$")
    password: str = Field(min_length=8)
    age: int = Field(ge=13, le=120)
    address: Address                    # nested

    @field_validator("tags")
    @classmethod
    def clean(cls, tags): return sorted({t.lower() for t in tags})

class UserOut(BaseModel):               # no password field!
    username: str
```

> Strings like `"30"` become the right types.

> Every error comes with its location: `address.country`.

> Separate **In** and **Out** models: secrets never leak.

LESSON · DEPENDENCIES

Depends(): reusable, testable building blocks

```python
def current_user(authorization: Annotated[str | None, Header()] = None):
    user = USERS.get((authorization or "").removeprefix("Bearer "))
    if user is None: raise HTTPException(401, "not logged in")
    return user

def admin_user(user = Depends(current_user)):      # chains
    if user["role"] != "admin": raise HTTPException(403)
    return user

@products.delete("/{id}", status_code=204)
def delete(id: int, admin: AdminDep): ...

app.dependency_overrides[current_user] = fake_user  # tests
```

> **APIRouter**: group endpoints, with a prefix such as `/api/v1`.

> Pagination, DB sessions, settings, and the current user are all dependencies.

LESSON · PRODUCTION PLUMBING

Errors, middleware, CORS, settings

FEATURE	WHAT IT GIVES YOU
@app.exception_handler	domain errors → clean JSON (409); no tracebacks leaked
middleware	request IDs, timing headers, access logs on every request
CORSMiddleware	only your frontends may call the API from a browser
pydantic-settings	env vars → typed config; `SecretStr` hides secrets
lifespan	open shared clients/pools at startup, close at shutdown
async def	for awaited I/O (httpx, async DB); plain `def` for blocking code

EXERCISE 22 · YOUR TURN

A library API

The task

1	APIRouter at /api/v1/books
2	author filter + limit/offset validation
3	X-API-Key dependency on POST (401)
4	BookNotFound → custom 404 JSON

```
GET /api/v1/books?author=martin
{"total": 2, "items": [...]}
GET /api/v1/books/9
404 {"error": "book_not_found", "id": 9}
```

Starter:
exercises/ex22_fastapi_library_api.py
Solution:
solutions/ex22_fastapi_library_api.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

23

SQLAlchemy 2.0 & Alembic

Databases through an ORM, without N+1, with versioned migrations.

IN THIS MODULE

Models	`models.py`
ORM CRUD	`01_orm_crud.py`
N+1 & loading	`02_n_plus_1_and_loading.py`
SQLAlchemy + FastAPI	`03_sqlalchemy_with_fastapi.py`
Alembic	`04_alembic_migrations.py`
Exercise: blog ORM	`ex23_blog_orm.py`

LESSON · THE ORM

Tables as typed classes

```python
class Book(Base):
    __tablename__ = "books"
    id: Mapped[int] = mapped_column(primary_key=True)
    title: Mapped[str] = mapped_column(String(200))
    author_id: Mapped[int] = mapped_column(ForeignKey("authors.id"))
    author: Mapped[Author] = relationship(back_populates="books")

with Session(engine) as session:
    cheap = session.scalars(select(Book).where(Book.price < 40)).all()
    book.price = 29.0          # just change it...
    session.commit()           # ...one transaction
```

> The same code runs on SQLite, PostgreSQL and **Azure SQL**.

> Values are bound parameters: **no SQL injection**.

LESSON · N+1 AND MIGRATIONS

The two things that bite in production

PROBLEM	FIX	MEASURED
lazy loading in a loop	`selectinload(Author.books)`	51 → **2** queries
many-to-one in a loop	`joinedload(Book.author)`	**1** query
schema changes on live DBs	**Alembic** migrations	data survives
	`alembic upgrade head`	apply missing steps
	`alembic downgrade -1`	undo the last step

```python
def get_db():
    db = SessionLocal()
    try:
        yield db          # one session per request
    finally:
        db.close()
```

EXERCISE 23 · YOUR TURN

A blog database

The task

1	User, Post, Tag + a many-to-many table
2	publish(): reuse existing tags
3	posts_with_tag, top_authors (GROUP BY)
4	all posts with tags in ≤ 2 queries

```
top_authors(s)
# [("vishal", 2), ("raj", 1)]
queries: 2   # not N+1
```

Starter: exercises/ex23_blog_orm.py
Solution: solutions/ex23_blog_orm.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

24

Authentication & security

Store passwords right, issue tokens, and stop the OWASP classics.

IN THIS MODULE

Password hashing	01_password_hashing.py
JWT tokens	02_jwt_tokens.py
FastAPI login	03_fastapi_login.py
Security checklist	04_web_security_checklist.py
Exercise: auth toolkit	ex24_auth_toolkit.py

LESSON · PASSWORDS

Slow + salted, never sha256

```python
def hash_password(pw):
    salt = secrets.token_bytes(16)            # unique per user
    digest = hashlib.scrypt(pw.encode(), salt=salt, n=2**14, r=8, p=1)
    return f"scrypt${salt.hex()}${digest.hex()}"

def verify_password(pw, stored):
    _, salt, digest = stored.split("$")
    check = hashlib.scrypt(pw.encode(), salt=bytes.fromhex(salt), n=2**14, r=8, p=1)
    return hmac.compare_digest(check.hex(), digest)  # constant time
```

> sha256: ~1,000,000 guesses/s on one core.
> scrypt: **50 ms** each.

> In production: **argon2id** (argon2-cffi / pwdlib).

> Tokens and reset links: `secrets`, never `random`.

LESSON · JWT + LOGIN

Signed, not secret

```
token = jwt.encode({"sub": "vishal", "role": "admin",
                    "exp": now + timedelta(minutes=15)},
                   SECRET, algorithm="HS256")

claims = jwt.decode(token, SECRET, algorithms=["HS256"])  # pin it!

@app.get("/admin")
def admin(user = Depends(require_role("admin"))): ...
#  no / bad token -> 401   wrong role -> 403
```

> The payload is **base64**: anyone can read it. Put no secrets in it.

> Tampering → `InvalidSignatureError`; expired → rejected.

> Same message for wrong user and wrong password.

LESSON · OWASP BASICS

A checklist you can run

THREAT	DEFENCE
SQL injection	parameters / ORM
XSS	escape output + Content-Security-Policy
Broken access control	check ownership on **every** request (401 vs 403)
Path traversal	`resolve()` + `is_relative_to(UPLOADS)`
Brute force	rate limiting → **429** + Retry-After
CSRF	SameSite cookies + CSRF tokens (Bearer headers aren't auto-sent)
Leaked secrets / vulnerable deps	env vars / Key Vault · `pip-audit`

EXERCISE 24 · YOUR TURN

An auth toolkit

The task

1	scrypt hash + verify
2	create/read JWT; AuthError on any bad token
3	POST /login, GET /orders (401)
4	DELETE /orders/{id}: admins only (403)

```
read_token(expired)  # AuthError
DELETE /orders/1 as raj    -> 403
DELETE /orders/1 as admin  -> 204
```

Starter: exercises/ex24_auth_toolkit.py
Solution: solutions/ex24_auth_toolkit.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

25

Django + DRF

Batteries included: ORM, admin, templates, auth, and a REST API.

IN THIS MODULE

Guided tour	`01_django_tour.py`
The project	`bookstore_project/`
The API	`catalog/api.py`
Tests	`catalog/tests.py`
Exercise: reviews	`ex25_test_reviews.py`

LESSON · DJANGO

Model → migration → admin → view → template

```
class Book(models.Model):
    title = models.CharField(max_length=200)
    author = models.ForeignKey(Author, on_delete=models.CASCADE,
                               related_name="books")
    price = models.DecimalField(max_digits=7, decimal_places=2)

$ python manage.py makemigrations && python manage.py migrate
$ python manage.py createsuperuser && python manage.py runserver

Book.objects.filter(price__lt=35).select_related("author")
```

admin.py: a full back office with search and filters, for free.

DecimalField for money, never float.

Settings from **env vars**; `DEBUG` off in production.

Serializers + ViewSets + a router

```python
class BookSerializer(serializers.ModelSerializer):
    author_name = serializers.CharField(source="author.name", read_only=True)
    class Meta:
        model = Book
        fields = ["id", "title", "author", "author_name", "price"]
    def validate_price(self, value):
        if value <= 0: raise serializers.ValidationError("price must be positive")
        return value

class BookViewSet(viewsets.ModelViewSet): ...     # list/create/retrieve/update/delete
router.register("books", BookViewSet, basename="book")
```

> **IsAuthenticatedOrReadOnly**: guests read, users write.

> Django when you need admin + auth + HTML; FastAPI for lean async APIs.

EXERCISE 25 · YOUR TURN

Book reviews

The task

1	Review model (1-5) + migration
2	admin + ReviewSerializer validation
3	avg_rating on books (Avg annotate)
4	/api/reviews/?book=

```
$ python manage.py test
Ran 8 tests ... OK
```

Starter: exercises/ex25_test_reviews.py
Solution: solutions/ex25_django_reviews/

> Done? Open a **Pull Request** and I'll review your code.

MODULE

26

WebSockets, tasks & caching

Push live updates, move slow work out of the request, cache with Redis.

IN THIS MODULE

WebSocket chat	`01_websockets_chat.py`
Background tasks	`02_background_tasks.py`
Caching with Redis	`03_caching_with_redis.py`
Exercise: live prices	`ex26_live_prices.py`

The server can push

```python
@app.websocket("/ws/{name}")
async def chat(ws: WebSocket, name: str):
    await room.join(ws, name)            # accept + remember
    try:
        while True:
            text = await ws.receive_text()
            await room.broadcast(f"{name}: {text}")
    except WebSocketDisconnect:
        await room.leave(ws)
```

> Starts as HTTP, **upgrades**, then stays open both ways.

> Chat, notifications, dashboards, live prices.

Answer fast, do the slow part later

TOOL	USE
BackgroundTasks	small jobs after the response (welcome email)
queue + worker	**202** + job id, then the client polls `/jobs/{id}`
Celery / RQ / arq	separate workers, retries, schedules, survive restarts
Redis cache-aside	read → miss → DB → `set(key, ex=60)`; write → `delete(key)`
Redis extras	counters (`incr`), leaderboards (`zadd`), rate limits

```
1st read:  899.0 EUR in 224.5 ms
2nd read:  899.0 EUR in   0.1 ms
after update: 799.0 EUR  <- fresh
```

EXERCISE 26 · YOUR TURN

Live prices

The task

1	GET /products/{id}: cache-aside, X-Cache HIT/MISS
2	30 s TTL, 404 when missing
3	PUT: update DB + delete cache key
4	push {id, price} to every /ws/prices client

```
X-Cache: MISS -> HIT
PUT /products/1?price=799.0
ws <- {"id": 1, "price": 799.0}
```

Starter: exercises/ex26_live_prices.py
Solution: solutions/ex26_live_prices.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

27

Docker, CI/CD & Azure

From laptop to cloud: containers, pipelines, Azure Container Apps.

IN THIS MODULE

Production-ready tour	`01_production_ready_tour.py`
Dockerfile	`Dockerfile`
docker compose	`docker-compose.yml`
CI/CD pipeline	`github-actions/ci-cd.yml`
Azure deploy	`azure/deploy.sh`
Exercise: fix a Dockerfile	`ex27_fix_the_dockerfile.py`

LESSON · DOCKERFILE

Small, cached, non-root

```
FROM python:3.12-slim AS build
COPY requirements.txt .
RUN pip install --no-cache-dir --prefix=/install         -r requirements.txt

FROM python:3.12-slim
RUN useradd --create-home --uid 1000 appuser
COPY --from=build /install /usr/local
COPY app ./app                  # code LAST: cached layers
USER appuser
HEALTHCHECK CMD python -c "...urlopen('.../health/live')"
CMD ["gunicorn", "app.main:app", "-k",
     "uvicorn.workers.UvicornWorker", "-b", "0.0.0.0:8000"]
```

12-factor: config from env, logs to stdout as JSON, health probes.

Liveness = restart me · **Readiness** = send me traffic.

Nothing ships unless the tests pass

```yaml
jobs:
  test:                          # every push + PR
    steps: [checkout, setup-python, ruff, mypy, pytest]
  deploy:
    needs: test                  # only if green
    if: github.ref == 'refs/heads/main'
    steps:
      - uses: azure/login@v2     # OIDC: no stored passwords
      - run: az acr build --image hello-service:${{ github.sha }} ...
      - run: az containerapp update --image ...:${{ github.sha }}
```

> Tag images with the **commit SHA**: you know what runs, and rollback is easy.

> Container Apps: HTTPS, **scale to zero**, secrets via `secretref:`.

> `docker compose up`: API + Postgres + Redis locally.

EXERCISE 27 · YOUR TURN

Fix a bad Dockerfile

The task

1	slim pinned base, no latest
2	deps before code, --no-cache-dir
3	non-root USER, no baked secrets
4	HEALTHCHECK, exec-form CMD, no --reload

```
Reviewed ex27_Dockerfile.bad: 0/9
Reviewed ex27_Dockerfile:     9/9
All checks passed!
```

Starter: exercises/ex27_fix_the_dockerfile.py
Solution: solutions/ex27_Dockerfile

> Done? Open a **Pull Request** and I'll review your code.

MODULE

28

Web capstone: taskboard

Auth, SQL, permissions, live updates, tests, Docker: all in one API.

IN THIS MODULE

The project	`taskboard/`
Architecture	`taskboard/README.md`
Tests	`taskboard/tests/`
Challenges	`ex28_web_capstone_challenges.md`
Final exam	`28_web_capstone/QUIZ.md`

CAPSTONE · ARCHITECTURE

Every web module in one service

FILE	JOB	MODULE
config.py	TASKBOARD_* env vars	22, 27
db.py · models.py	engine, session per request, User 1---< Task	23
schemas.py	pydantic in/out, validation	22
security.py · deps.py	scrypt, JWT, current_user (401)	24
routers/tasks.py	CRUD, owner-or-admin (403), pagination	22-24
realtime.py	WebSocket push on every change	26
Dockerfile	multi-stage, non-root, health check	27

```
$ pytest      7 passed
$ mypy        Success (strict)
$ ruff check  All checks passed!
```

EXERCISE 28 · YOUR TURN

Web capstone challenges

The task

1	⭐ due dates + an admin view
2	⭐ ⭐ Alembic · login rate limit · Redis counts
3	⭐ ⭐ ⭐ projects with members, no N+1
4	⭐ ⭐ ⭐ per-user WebSocket · deploy to Azure

```
# for EVERY challenge:
# 1. write the test first
# 2. make it pass
# 3. keep pytest, mypy, ruff green
```

Starter:
exercises/ex28_web_capstone_challenges.md

> Done? Open a **Pull Request** and I'll review your code.

BONUS · RECURSION, DEBUGGING, TEXT & BYTES

Three skills every pro needs

LESSON	KEY IDEA
10 · 07_recursion.py	base case + smaller problem; the call stack; `@lru_cache` turns fib(30) from 2.7M calls into 31
13 · 06_debugging_like_a_pro.py	`breakpoint()`, pdb commands, post-mortem, `faulthandler`, VS Code launch.json
18 · 07_text_and_bytes.py	`str` is text, `bytes` is data; always `encoding="utf-8"`; mojibake; NFC

```
def total_size(folder):
    if folder.is_file():          # base case
        return folder.size
    return sum(total_size(c)      # smaller problem
               for c in folder.children)

(Pdb) n  s  c  p total  w  q
```

MODULE

29

Algorithms & data structures

The classics behind every coding interview, timed so you see Big-O.

IN THIS MODULE

Sorting & searching	`01_sorting_and_searching.py`
Stacks, queues, linked lists	`02_stacks_queues_linked_lists.py`
Trees	`03_trees.py`
Graphs	`04_graphs.py`
Dynamic programming	`05_dynamic_programming.py`
Problem-solving patterns	`06_problem_solving_patterns.py`
Exercise: interview problems	`ex29_interview_problems.py`

LESSONS 01-03 · CHEAT SHEET

Pick the algorithm by its Big-O

ALGORITHM / STRUCTURE	TIME	REMEMBER
bubble / insertion sort	O(n²)	fine for tiny or almost-sorted data
merge sort · `sorted()`	O(n log n)	stable; Python's Timsort is merge-based
quick sort	O(n log n) avg	O(n²) on a bad pivot
binary search	O(log n)	data must be **sorted**; `bisect`
stack (list) · **queue** (deque)	O(1) push/pop	undo, brackets · BFS, jobs
BST	O(log n) … O(n)	in-order = sorted; sorted inserts make it a list

Measure, don't guess: the lesson times bubble vs merge sort on 2,000 numbers.

Dijkstra: cheapest route with a heap

```python
def dijkstra(graph, start):
    best = {start: 0}
    heap = [(0, start)]                    # (distance, city)
    while heap:
        dist, city = heapq.heappop(heap)   # closest unfinished city
        if dist > best.get(city, inf):
            continue                       # outdated entry
        for nxt, km in graph[city].items():
            if dist + km < best.get(nxt, inf):
                best[nxt] = dist + km
                heapq.heappush(heap, (dist + km, nxt))
    return best
```

BFS (queue): fewest hops when every edge counts the same.

DFS (stack): explore everything, find cycles.

Topological sort: a valid order for tasks with dependencies.

LESSONS 05-06 · PATTERNS

Most interview problems are one of these

PATTERN	SIGNAL IN THE QUESTION	EXAMPLE
Hash map	”have I seen …?”, pairs, counts	two sum O(n), anagrams
Two pointers	sorted data, both ends	palindrome, pair sum
Sliding window	longest / best sub-array or substring	longest unique substring
Prefix sums	many range-sum questions	sales of days i..j in O(1)
BFS / DFS	grid, network, connected parts	count islands
Dynamic programming	overlapping subproblems, ”fewest / most ways”	coin change, edit distance

Process: restate → brute force → find the pattern → code → edge cases → state Big-O.

EXERCISE 29 · YOUR TURN

Six classic interview problems

The task

1	two_sum in O(n)
2	valid_brackets with a stack
3	merge_intervals · longest_unique_substring
4	count_islands (BFS) · min_coins (DP)

```
$ python exercises/ex29_interview_problems.py
AssertionError            # until you solve it

$ python solutions/ex29_interview_problems.py
All checks passed!        # incl. a 1 s time limit
```

Starter:
exercises/ex29_interview_problems.py
Solution:
solutions/ex29_interview_problems.py

> Done? Open a **Pull Request** and I'll review your code.

MODULE

30

AI with Python

LLM APIs, tools, RAG and agents behind FastAPI: safe, cheap and tested.

IN THIS MODULE

Shared client (offline or real)	`llm_client.py`
LLM API basics	`01_llm_api_basics.py`
Structured output & tools	`02_structured_output_and_tools.py`
Embeddings & RAG	`03_embeddings_and_rag.py`
Agent behind an API	`04_agent_api.py`
Safety, cost & evals	`05_safety_cost_eval.py`
Exercise: AI toolkit	`ex30_ai_toolkit.py`

LESSON 01 · LLM API BASICS

One endpoint: messages in, a message out

```python
client = anthropic.Anthropic()          # reads ANTHROPIC_API_KEY
response = client.messages.create(
    model="claude-opus-5-5",
    max_tokens=1000,
    system="You are a concise Python tutor.",
    messages=[{"role": "user", "content": "What is a generator?"}],
)
if response.stop_reason == "max_tokens": ...   # cut off!
text = "".join(b.text for b in response.content if b.type == "text")
print(response.usage.input_tokens, response.usage.output_tokens)

with client.messages.stream(...) as stream:     # word by word
    for chunk in stream.text_stream: print(chunk, end="")
```

> The API is **stateless**: resend the history every turn.

> `content` is a list of blocks: text, thinking, tool_use.

> Steer with the **system prompt** and `effort`; current models reject `temperature`.

Every lesson runs **offline** with a fake client. `USE_REAL_LLM=1` switches to Claude (or Azure Foundry).

LESSON 02 · TOOLS

The agent loop: the model asks, your code acts

```
messages = [{"role": "user", "content": question}]
for step in range(max_steps):
    r = client.messages.create(model=MODEL, max_tokens=4000,
                               tools=TOOLS, messages=messages)
    if r.stop_reason != "tool_use":
        return text_of(r)                       # final answer
    messages.append({"role": "assistant", "content": r.content})
    results = []
    for b in r.content:
        if b.type == "tool_use":
            try:    out = REGISTRY[b.name](**b.input)
            except Exception as e:
                out, err = f"error: {e}", True   # is_error=True
            results.append({"type": "tool_result",
                            "tool_use_id": b.id, "content": str(out)})
    messages.append({"role": "user", "content": results})
```

Tools = name + description + **JSON schema**; `strict: true` keeps arguments valid.

Structured output: `client.messages.parse(output_format=Ticket)` → a validated pydantic object.

Send **all** tool results back in **one** message.

Answer from YOUR documents, with sources

STEP	WHAT HAPPENS	IN THE LESSON
1 Chunk	split docs into small overlapping pieces	2-sentence windows
2 Embed	text → vector of numbers	TF-IDF with numpy (real: Voyage, OpenAI…)
3 Store	keep the vectors	a matrix (prod: pgvector, Azure AI Search)
4 Retrieve	cosine similarity, top-k	`matrix @ embed(q)`
5 Generate	send ONLY those chunks + the question	”cite [returns.md] or say I don’t know”

No relevant chunk? Answer ”I don’t know” **without** calling the LLM: cheaper and honest.

FastAPI + the LLM as a dependency

```python
class AskIn(BaseModel):
    question: str = Field(min_length=3, max_length=500)

def get_llm():
    return get_client()              # real SDK in production
LLM = Annotated[Any, Depends(get_llm)]

@app.post("/ask", response_model=AskOut)
def ask(body: AskIn, llm: LLM):
    return run_agent(llm, body.question)   # max_steps, tokens used

# tests: fast, free, repeatable
app.dependency_overrides[get_llm] = lambda: FakeClient(...)
```

> Limit input size and **agent steps**; return 504 if it loops.

> Report **token usage** per request: log it, bill it.

> `/ask/stream`: `StreamingResponse` over `stream.text_stream`.

LESSON 05 · SHIP IT SAFELY

Injection, privacy, cost, evals

RISK	DO THIS
Prompt injection	least privilege tools; limits & approvals in **code**; untrusted text in tags = data
Privacy	redact emails, IBANs, phones **before** the API call
Cost	tokens in × price + tokens out × price; budgets; smaller model for easy routes; caching; Batch API
Quality	an **eval set**: questions + checks, run on every prompt or model change
Refusals	check `stop_reason == "refusal"`; server-side fallbacks

```
3,000 in + 400 out, 10k chats/day, 30 days
opus-5-5   0.0200 USD/chat -> 6,000 USD/month
sonnet-5-5 0.0100 USD/chat -> 3,000 USD/month
haiku-5-5  0.0005 USD/chat ->   150 USD/month
```

EXERCISE 30 · YOUR TURN

Your AI toolkit

The task

1	chunk_words · cosine · top_k
2	cost_usd from a price table
3	redact emails and phone numbers
4	run_tool_loop: errors, all results in one message, max_steps

```
$ python exercises/ex30_ai_toolkit.py
AssertionError            # until you solve it

$ python solutions/ex30_ai_toolkit.py
All checks passed!        # no API key needed
```

Starter: exercises/ex30_ai_toolkit.py
Solution: solutions/ex30_ai_toolkit.py

> Done? Open a **Pull Request** and I'll review your code.



YOU MADE IT

From Foundations to Pro, the Web and AI

Take the **final exams** in 20_capstone and 28_web_capstone, deploy taskboard, build an AI agent, and share it on LinkedIn.

- youtube.com/@CodeMitVishal
- github.com/CodeMitVishal/python-pro
- buymeacoffee.com/codemitvishal